Worksheet 1.1 — Power Functions & End Behaviour

Grade 12 Advanced Functions (MHF4U) · Unit 1: Polynomial Functions

The Big Ideas

Every polynomial behaves like its leading term at the ends. Degree parity and the sign of a decide the end behaviour.

- Even degree: both ends the same (upup if $a > 0$, downdown if $a < 0$).
- Odd degree: ends opposite (downup if $a > 0$, updown if $a < 0$).
- Write it with limits, e.g. as $x \to +\infty,\ y \to +\infty$.

Core Concepts

Power functions

A **power function** has the form $y = ax^n$ with n a whole number; its degree and the sign of a dictate the overall shape.

End behaviour

As $x \to \pm\infty$, the ends are set by the degree's parity and the sign of a: even degrees send both ends the same way, odd degrees send them opposite ways.

Even vs odd symmetry

Even-degree power functions are symmetric about the y-axis; odd-degree ones have rotational symmetry about the origin.

Worked Examples

Example 1 — Odd, positive

End behaviour of $f(x) = x^3$?

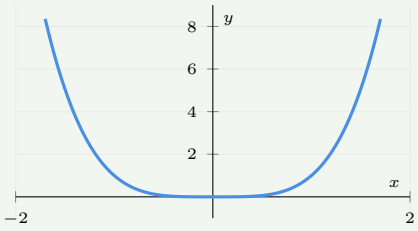
Solution. Odd degree, $a > 0$: as $x \to -\infty,\ y \to -\infty$; as $x \to +\infty,\ y \to +\infty$ (downup):



Example 2 — Even, positive

End behaviour of $f(x) = x^4$?

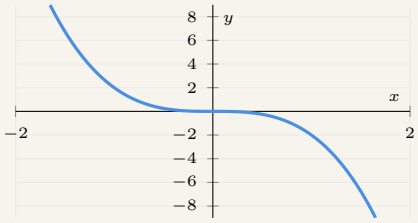
Solution. Even degree, $a > 0$: both ends $y \to +\infty$ (upup):



Example 3 — Odd, negative

End behaviour of $f(x) = -2x^3$?

Solution. Odd, $a < 0$ flips it: as $x \to -\infty,\ y \to +\infty$; as $x \to +\infty,\ y \to -\infty$ (updown):



Example 4 — Even, negative

End behaviour of $f(x) = -x^4$?

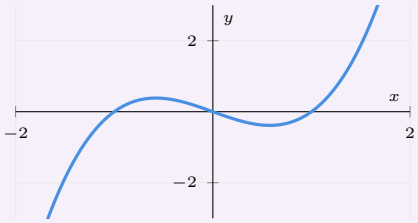
Solution. Even, $a < 0$: both ends $y \to -\infty$ (downdown):



Example 5 — Read a graph

A graph falls on the left and rises on the right.

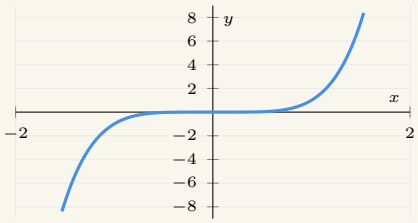
Solution. Opposite ends odd degree; rising on the right $a > 0$ (e.g. $y = x^3 - x$):



Example 6 — Odd, positive

End behaviour of $f(x) = x^5$?

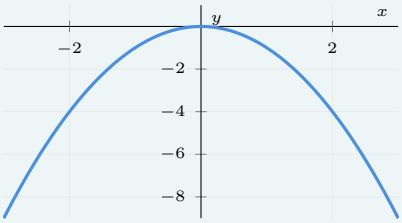
Solution. Downup (odd, $a > 0$):



Example 7 — Even, negative

End behaviour of $f(x) = -x^2$?

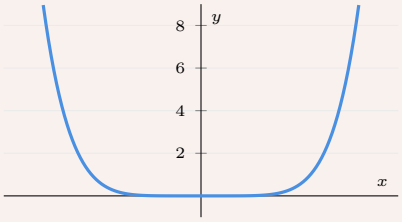
Solution. Downdown (even, $a < 0$):



Example 8 — Even, positive

End behaviour of $f(x) = 3x^6$?

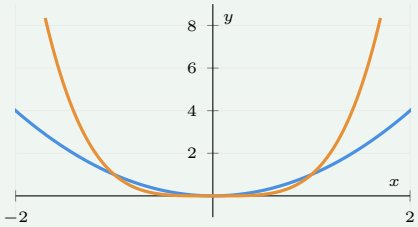
Solution. Upup (even, $a > 0$):



Example 9 — Compare shapes

How do $y = x^2$ and $y = x^4$ compare?

Solution. Both upup; x^4 is flatter near 0 and steeper outside $[-1, 1]$:



Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

End behaviour of $f(x) = x^7$?

Question 2

End behaviour of $f(x) = -x^6$?

Question 3

End behaviour of $f(x) = 2x^4$?

Question 4

End behaviour of $f(x) = -5x^3$?

Question 5

A graph rises on both ends. Even or odd degree?

Question 6

End behaviour of $f(x) = 0.5x^7$?

Question 7

End behaviour of $f(x) = -x^5$?

Question 8

End behaviour of $f(x) = 4x^2$?

Question 9

A graph falls on both ends. Sign of a?

Question 10

End behaviour of $f(x) = -2x^6$?

Question 11

Does $f(x) = 2x^{10} + x$ open up or down?

Question 12

A cubic with $a < 0$: describe its ends.

Question 13 — Challenge

A polynomial has ends updown. State the degree parity and sign of a.

Question Recap & Answer Key

Question Recap

1. End behaviour of $f(x) = x^7$?
2. End behaviour of $f(x) = -x^6$?
3. End behaviour of $f(x) = 2x^4$?
4. End behaviour of $f(x) = -5x^3$?
5. A graph rises on both ends. Even or odd degree?
6. End behaviour of $f(x) = 0.5x^7$?
7. End behaviour of $f(x) = -x^5$?
8. End behaviour of $f(x) = 4x^2$?
9. A graph falls on both ends. Sign of a?
10. End behaviour of $f(x) = -2x^6$?
11. Does $f(x) = 2x^{10} + x$ open up or down?
12. A cubic with $a < 0$: describe its ends.
13. A polynomial has ends updown. State the degree parity and sign of a.

✓ Answer Key

1. downup
2. downdown
3. upup
4. updown
5. even
6. downup
7. updown
8. upup
9. $a < 0$
10. downdown
11. up
12. updown
13. odd, $a < 0$